Workshop: Imperial Units and Calculating Body Indices

UEA LEARNING ENANCEMENT TEAM – MATHEMATICS & STATISTICS

Imperial and Metric Units for Volumes

For this section you should use:

$$1 \text{ pint } \approx 500 \text{ ml}$$
$$1 \text{ gallon} = 8 \text{ pints}$$

1. Convert the following volumes into ml:

   a. 3 pints

   $$3 \times 500 = 1500\ ml$$

   b. 0.5 pints

   $$0.5 \times 500 = 250\ ml$$

   c. 3 gallons

   $$3 \times 8 = 24\ pints$$
   $$24 \times 500 = 12000\ ml$$

2. Convert the following volumes into pints:

   a. 2000 ml

   $$2000 \div 500 = 4\ pints$$

   b. 3.5 l

   $$3500 \div 500 = 7\ pints$$

   c. 4 gallons

   $$4 \times 8 = 32\ pints$$

   d. 125 ml

   $$125 \div 500 = 0.25\ pints$$

Imperial and Metric Units for Mass

For this section you should use:

$$1 \text{ kg} \approx 2 \text{ lbs (pounds)}$$
$$1 \text{ lb} = 16 \text{ oz (ounces)}$$
$$1 \text{ stone} = 14 \text{ lbs}$$

3. Convert the following masses into lbs (pounds):

   a. 5 kg

   $$5 \times 2 = 10\ lbs$$

   b. 500 g

   $$0.5 \times 2 = 1\ lbs$$

   c. 3500 g

   $$3.5 \times 2 = 7\ lbs$$

   d. 32 oz

   $$32 \div 16 = 2\ lbs$$

   e. 8 oz

   $$8 \div 16 = 0.5\ lbs$$

   f. 2 stone

   $$2 \times 14 = 28\ lbs$$

   g. 10 stone 4 lbs

   $$10 \times 14 + 4 = 140 + 4 = 144\ lbs$$

   h. 12 stone 12 lbs

   $$12 \times 14 + 12 = 180\ lbs$$

4. Convert the following into stone:

   a. 280 lbs

   $$280 \div 14 = 20\ stone$$

   b. 84 lbs

   $$84 \div 14 = 6\ stone$$

   c. 70 kg

   $$70 \times 2 = 140\ lbs$$
   $$140 \div 14 = 10\ stone$$

5. Convert the following masses into stone and lbs (e.g. 12 stone 5 lbs):

a. 150 lbs

$$150 \div 14 = 10\,r\,10 = 10\ stone\ 10\ lbs$$

b. 200 lbs

$$200 \div 14 = 14\,r\,4 = 14\ stone\ 4\ lbs$$

c. 55 kg

$$55 \times 2 = 110\ lbs$$
$$110 \div 14 = 7\,r\,12 = 7\ stone\ 12\ lbs$$

d. 82 kg

$$82 \times 2 = 164\ lbs$$
$$164 \div 14 = 11\ stone\ 10\ lbs$$

Imperial and Metric Units for Length

For this section you should use:

$$1 \text{ inch } \approx 2.5 \text{ cm}$$
$$1 \text{ ft (foot)} = 12 \text{ inches}$$

6. Convert the following lengths into cm:

a. 10 inches

$$10 \times 2.5 = 25\ cm$$

b. 1 ft

$$1 \times 12 = 12\ inches$$
$$12 \times 25 = 30\ cm$$

c. 5 ft 3 inches

$$5 \times 12 + 3 = 63\ inches$$
$$63 \times 2.5 = 157.5\ cm$$

d. 6 ft 6 inches

$$6 \times 12 + 6 = 78\ inches$$
$$78 \times 2.5 = 195\ cm$$

7. Convert the following lengths into ft and inches (e.g. 5 ft 8 inches):

a. 240 inches

$$240 \div 12 = 20\ ft$$

b. 100 inches

$$100 \div 12 = 8\,r\,4 = 8\ ft\ 4\ inches$$

c. 75 inches

$$75 \div 12 = 6\,r\,3 = 6\,ft\,3\,inches$$

d. 50 cm

$$50 \div 2.5 = 20\,inches$$
$$20 \div 12 = 1\,r\,8 = 1\,ft\,8\,inches$$

e. 2 m

$$200 \div 2.5 = 80\,inches$$
$$80 \div 12 = 6\,r\,8 = 6\,feet\,8\,inches$$

Body Mass Index

The formula for body mass index (BMI):

$$\text{BMI} = \frac{\text{Mass (kg)}}{(\text{Height (m)})^2}$$

8. Find the BMI of the following people:

a. Mass = 90 kg, Height = 2 m

$$\frac{90}{2^2} = \frac{90}{4} = 22.5\ kg/m^2$$

b. Mass = 64 kg, Height = 1.6 m

$$\frac{64}{1.6^2} = \frac{64}{2.56} = \frac{6400}{256} = 25\ kg/m^2$$

c. Mass = 64.8 kg, Height = 180 cm

$$\frac{64.8}{1.8^2} = \frac{64.8}{3.24} = \frac{6480}{324} = 20\ kg/m^2$$

Body Surface Area

The formula for body surface area (BSA):

$$\text{BSA } (\text{m}^2) = \sqrt{\frac{\text{Height (cm)} \times \text{Mass (Kg)}}{3600}}$$

The following information may be useful:

$$\sqrt{2} \approx 1.41 \quad \sqrt{3} \approx 1.73 \quad \sqrt{5} \approx 2.24 \quad \sqrt{6} \approx 2.45$$

9. Find the BSA of the following people:
   a. Height = 180 cm, Mass = 60 kg

$$BSA = \sqrt{\frac{180 \times 60}{3600}} = \sqrt{\frac{18 \times 6}{36}} = \sqrt{\frac{1 \times 6}{2}} = \sqrt{3} \approx 1.73\ m^2$$

   b. Height = 144 cm, Mass = 50 kg

$$BSA = \sqrt{\frac{144 \times 50}{3600}} = \sqrt{\frac{144 \times 5}{360}} = \sqrt{\frac{144 \times 1}{72}} = \sqrt{2} \approx 1.41\ m^2$$

   c. Height = 1.92 m, Mass = 112.5 kg

$$BSA = \sqrt{\frac{192 \times 112.5}{3600}} = \sqrt{\frac{4 \times 112.5}{75}} = \sqrt{\frac{450}{75}} = \sqrt{6} \approx 2.45\ m^2$$

Ideal Body Weight (IBW)

The formula for IBW is:

$$\text{Female: IBW (kg)} = \text{Height (cm)} \times 0.9 - 92$$
$$\text{Male: IBW (kg)} = \text{Height (cm)} \times 0.9 - 88$$

10. What is the IBW for the following:

   a. Male, Height = 1.7 m

$$IBW = 170 \times 0.9 - 88 = 153 - 88 = 65\ kg$$

   b. Female, Height = 1.8 m

$$IBW = 180 \times 0.9 - 92 = 162 - 92 = 70\ kg$$

Mixed Question

11. For a person who is:

Male; Height = 6 ft; Mass = 14 stone 4 lbs

Find:

a. BSA
b. IBW

$$\text{Height} = 6 \times 12 = 72\ inches$$
$$72 \times 2.5 = 180\ cm$$

$$Mass = 14 \times 14 + 4 = 200\ lbs$$
$$200 \div 2 = 100\ kg$$

a. $BSA = \sqrt{\frac{180 \times 100}{3600}} = \sqrt{\frac{180 \times 1}{36}} = \sqrt{5} \approx 2.24\ m^2$

b. $IBW = 180 \times 0.8 - 88 = 74\ kg$